

Mano Estrada

10/1/20

Individual planning: project concept and partnership line not final.

1. I have to build a device that runs on a controller like Arduino or a logic chip. It needs to work in 5 areas which are display, user input, a sensor, a moving part, and programming. I have to test it, show it working, with all github documentation, poster, and cited code.

A strong project is one where every part actively depends on others. For example reading what the code decide something, the code makes a note. A weak project is 5 things working with nothing to do with each other.

Signature: Mano Estrada

Date: 10/1

Team Members:

Witness: Gabriel Amellaga

Date: 10/2

N/A

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2. Requirements & Constraints

- Controllers: Arduino, Raspberry Pi, Esp 8266, Logic Gates, and similar boards, No Vex
- 5 categories: Display, input, sensor, ~~mechanism~~ mechanism, and logic control
- Integration: The device has to work as one system, and has to work all the time.
- Limits: Lab parts are free. I got 250% fulfillment for the whole project. I can spend \$20 on parts
- Style & Looks: I can lose points for messy wiring, flimsy builds, pinch points, or shock risks.
- Scoring: Each category is worth up to 50 points. Higher scores go to work with need effort
- Checkpoints: Proposal (Jan 27), Status (Feb 6, 27) System Demo (Mar 26), Documentation Setup (April)
- Deliverables: Git/Hub Repo, notebook, demo
- Sources: I can use ~~code~~ code and circuits from internet

Signature: Mmmu EschodaDate: 10/11/20

Team Members:

Witness: 6adwel ArellanoDate: 10/12N/A

3. Personal Readiness

What I have:

- Skill - Arduino ~~Don~~ foundation
- Skill - cad/soldering/3D printing
- Skill - circuits foundation

What I need to work on

- soldering neat joints to not lose points
- time organization
- Research Method.

4.

	Display	input	Sensor	Hardware	Log.c	Risk
Pill dispenser for forgetful people	Led & Buzzer	Buttons & a keypad for PIN	IR sensor & limit switch	Stepper motor	Schedx code	Jamming
Solar tracker that follows sun	Oled showing angles	Potentiometer for sensitivity	light sensor & limit switch	2 motor tilt frame	data log	J. Frying motors
color sorter for small items	LED with counter	start / stop button	color sensor	Servo motor	Hue values	Color readings for name.

Signature: Mario EstradaDate: 10/1

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Witness: Gabriel AmallasDate: 10/2N/A

5. Current front-runner

I'm ~~going~~ thinking about the pill dispenser. It solves a real problem, and every part depends on the others: the clock triggers the motor, the sensor can know the pill was taken, and the green and blue keep user updated. It should be cheap, safe, and include real research on Steppers, clock module, & scanning data. The solar tracker seems too simple, and the color sensor depends too much on lighting.

6. Anticipated challenges

1. Challenge: Getting the carousel to drop due pill every time

Why: If it's unreliable, our ~~score~~ score will drop

Early test: Build a small test carousel to test 50 times

2. Challenge: Keeping the right time and maintaining the schedule

Why: A dispenser that forgets the time is useless
test: Use a ~~clock~~ clock module with backup battery

Signature: Mario Estrada

Date: 10/1/10

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Witness: Gabriel Arellano

Date: 10/2

N/A

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3. Creating: Making the logic gates truly useful, instead of added bonus.

Why? The bonus could be worth 10% but the professor said that my integrals were.

Early Test: Build a gate circuit in breadboard but stops the timer when the LED is on then try it in a breadboard.

7. Individual or Partner project

I am going to choose to working with a partner. Even though an unrelated partner won't get me a grade adjustment, I will team up with someone that meets deadline.

8. Questions & Clarifications

1. Do failed printouts count as used fillament?
2. How do I get a purchased approved?
3. Does purchase cost include shipping?
4. Does the project need to run on batteries?
or B will adapter allowed.

Signature: Maria Estrada

Date: 10/1

Team Members:

Witness: Gabriel Ameliga

Date: 10/2

NA

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9. Preparation Plan

1. Wire up Stepper motor & save code
2. test a clock module after unplugging it,
3. Draw logic gate circuit on tinkercad
4. Print and test ~~area~~ Carousel
5. Study 2 CSU projects
6. Outline proposal before Jan 22.

10. Final Takeaway

The most important thing is to pick a project that works reliably in all 5 categories together. A simple project that works every time will always beat a flashy one.

Assistance Acknowledgement

I used AI to help me come up with project concepts & how I would create it.

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Team Members:

N/AWitness: gabriel AmellessDate: 10/2Continued From Page # 18Continued On Page # N/A